# Project Proposal

Gibbs Sampler

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## 1. Motivation and Background

A central challenge in Bayesian statistics is the inability to sample directly from a complex, high-dimensional joint posterior distribution $P(\theta_1, \theta_2, \dots, \theta_d|X)$. When parameters are interdependent, analytical integration becomes impossible, and simple Monte Carlo methods fail due to the "curse of dimensionality."

The **Gibbs Sampler** addresses this problem by treating a multivariate distribution as a series of univariate problems. By iteratively sampling from the **full conditional distributions**, where each parameter is sampled given the most recent values of all others, the algorithm constructs a Markov Chain that converges to the true joint distribution. This makes it an essential computational engine for models with latent (hidden) variables.

## 2. Precise Research Statement

**Research Question:** *How does the Gibbs Sampler function as a computational tool to facilitate Bayesian inference in complex models, and how can it be extended to reliably detect latent regime changes in financial time series?*

**What we will demonstrate:**

- That the Gibbs Sampler correctly recovers known parameters under controlled simulation.
- That convergence diagnostics (trace plots, autocorrelation) can reliably identify burn-in.
- That the sampler can detect historically documented structural breaks (e.g., 2008 Financial Crisis, 2020 COVID-19 shock) in S&P 500 and Nifty 50 return series.

## 3. Technical Details

### 3.1 The Gibbs Sampling Algorithm

Given a joint distribution $P(\theta_1, \theta_2, \dots, \theta_d \mid X)$, the Gibbs Sampler generates samples by iterating through the full conditional distributions. At iteration $k+1$:

$$\theta_i^{(k+1)} \sim P\Big(\theta_i \;\Big|\; \theta_1^{(k+1)}, \dots, \theta_{i-1}^{(k+1)}, \theta_{i+1}^{(k)}, \dots, \theta_d^{(k)}\Big), \quad i = 1, \dots, d \tag{1}$$

Under mild regularity conditions (irreducibility and aperiodicity of the induced Markov Chain), the sequence $\{\theta^{(k)}\}_{k=1}^{\infty}$ converges in distribution to the target joint posterior.

### 3.2 Markov Switching Model (Hamilton, 1989)

The core financial application models a time series $y_t$ (log returns) as:

$$y_t = \mu_{S_t} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,\, \sigma^2_{S_t}) \tag{2}$$

where $S_t \in \{0, 1\}$ is a latent state (e.g., $S_t = 0$: Bear market, $S_t = 1$: Bull market) governed by a first-order Markov chain with transition matrix:

$$\mathbf{P} = \begin{pmatrix} p_{00} & 1 - p_{00} \\ 1 - p_{11} & p_{11} \end{pmatrix} \tag{3}$$

### 3.3 Full Conditional Distributions

Placing conjugate priors — $\mu_j \sim \mathcal{N}(\mu_0, \tau^2)$, $\sigma_j^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$, and $p_{jj} \sim \text{Beta}(a_0, b_0)$ — the full conditional posteriors required for Gibbs Sampling are:

$$\mu_j \mid \mathbf{y}, \mathbf{S}, \sigma_j^2 \sim \mathcal{N}\left(\frac{\tau^{-2}\mu_0 + \sigma_j^{-2}\sum_{t:S_t=j} y_t}{\tau^{-2} + n_j\sigma_j^{-2}},\; \left(\tau^{-2} + n_j\sigma_j^{-2}\right)^{-1}\right) \tag{4}$$

$$\sigma_j^2 \mid \mathbf{y}, \mathbf{S}, \mu_j \sim \text{Inv-Gamma}\left(\alpha_0 + \tfrac{n_j}{2},\; \beta_0 + \tfrac{1}{2}\sum_{t:S_t=j}(y_t - \mu_j)^2\right) \tag{5}$$

$$p_{jj} \mid \mathbf{S} \sim \text{Beta}(a_0 + n_{jj},\; b_0 + n_{j,1-j}) \tag{6}$$

where $n_j = |\{t : S_t = j\}|$ is the number of observations in regime $j$, and $n_{jj}$ is the number of transitions from state $j$ to state $j$.

### 3.4 Latent State Sampling via Forward-Backward Algorithm

The latent state sequence $\mathbf{S} = (S_1, \ldots, S_T)$ is sampled jointly using the *forward-filtering backward-sampling* (FFBS) algorithm (Carter & Kohn, 1994; Chib, 1996):

1. **Forward pass:** Compute filtered probabilities $P(S_t \mid y_1, \ldots, y_t)$ recursively using the prediction and update equations of the HMM filter.
2. **Backward sampling:** Draw $S_T$ from the filtered distribution at $T$, then sample $S_t \mid S_{t+1}, y_{1:t}$ backward for $t = T-1, \ldots, 1$.

### 3.5 Convergence Diagnostics

- **Trace plots:** Visual inspection of sampler trajectories to identify mixing and burn-in period.
- **Autocorrelation Function (ACF):** Slow decay indicates high autocorrelation and poor mixing.

- **Gelman-Rubin $\hat{R}$ statistic:** Compares within-chain and between-chain variance across multiple chains; $\hat{R} \approx 1$ indicates convergence.
- **Effective Sample Size (ESS):** Accounts for autocorrelation to assess the number of independent draws.

### 3.6 Computational Toolkit

| **Task** | **Tool / Library** |
|---|---|
| Core implementation | Python 3.x (NumPy, SciPy) |
| Financial data retrieval | `yfinance`, `pandas_datareader` |
| Bayesian diagnostics | `ArviZ` (trace plots, $\hat{R}$, ESS) |
| Visualization | `Matplotlib`, `Seaborn` |
| Statistical distributions | `scipy.stats` |

**Assumptions:**

- Return series are conditionally Gaussian within each regime.
- Regimes follow a first-order, stationary Markov process.
- Conjugate priors are sufficiently diffuse to let the data dominate.

## 4. Literature Review

The evolution of the Gibbs Sampler traces a path from image processing to its current status as a cornerstone of Bayesian econometrics and regime-switching models.

### 4.1 Foundations and Popularization

**Stochastic Relaxation, Gibbs Distributions, and the Bayesian Restoration of Images**
*Stuart Geman and Donald Geman (1984)*
The Gemans introduced the **Gibbs Sampler** as a breakthrough for sampling from complex joint probability distributions. By decomposing high-dimensional distributions into univariate conditionals, they proved that iterative sampling generates a Markov Chain converging to the true joint posterior. This established the theoretical foundation for MCMC methods used in this project.

**Sampling-Based Approaches to Calculating Marginal Densities**
*Alan E. Gelfand and Adrian F. M. Smith (1990)*
Gelfand and Smith popularized the sampler within the statistical community by demonstrating its efficiency in approximating marginal posteriors where analytical integration is impossible. Their work underscores the feasibility of our Case 2 and Case 3 implementations.

### 4.2 Latent Variables and Data Augmentation

**The Calculation of Posterior Distributions by Data Augmentation**
*Martin A. Tanner and Wing H. Wong (1987)*
Tanner and Wong introduced **Data Augmentation**, treating hidden states as "missing data." By sampling these latent variables alongside model parameters, they simplified

complex posteriors into tractable conditionals. This is the core mechanic used in our project to estimate unobserved market states.

### 4.3 Regime-Switching and Financial Applications

**A New Approach to the Economic Analysis of Nonstationary Time Series**
*James D. Hamilton (1989)*
Hamilton revolutionized time-series analysis with the **Markov Switching Model (MSM)**. He modeled non-stationary series as processes switching between distinct regimes (e.g., Bull vs. Bear) governed by a hidden Markov chain. This framework provides the structural basis for our financial application.

**Calculating Posterior Distributions in Markov Mixture Models**
*Siddhartha Chib (1996)*
Chib refined the Bayesian estimation of MSM by providing a systematic procedure for estimating transition probabilities. His *Forward-Filtering Backward-Sampling* (FFBS) algorithm is implemented in our Case 3 to improve sampling efficiency and convergence.

### 4.4 Modern Extensions and Convergence Diagnostics

**Bayesian Analysis of Time-Varying Parameter VAR**
*Nakajima & Watanabe (2020)*
This study demonstrates the use of contemporary diagnostics such as **Effective Sample Size (ESS)** and the $\hat{R}$ **statistic**. It serves as a modern template for our implementation of convergence checks to ensure the reliability of our financial forecasts.

**Generalized Mixtures and Telescoping Sampling**
*Sylvia Frühwirth-Schnatter (2023)*
This recent work introduces "telescoping" Gibbs Sampling, which adaptively determines the number of mixture components. While our project focuses on a two-state model, this paper informs our discussion on the limitations of fixed-regime implementations in shifting market environments.

## 5. Methodology: Study Plan

We will study and implement the Gibbs Sampler through the following three cases:

1. **Case 1: Multivariate Normal Recovery:** Simulate data from a bivariate normal with known parameters and high inter-variable correlation ($\rho = 0.9$). Apply the Gibbs Sampler and verify that posterior means recover the true parameters. Examine trace plots and ACF to study mixing as a function of $\rho$.

2. **Case 2: Bayesian Linear Regression:** Estimate regression coefficients $\boldsymbol{\beta}$ and error variance $\sigma^2$ under conjugate normal-inverse-gamma priors. Compare posterior means to OLS estimates as a sanity check. Demonstrate the full posterior distribution of each coefficient.

3. **Case 3: Latent Regime Detection (Financial Application):** Fit a two-state Markov Switching Model to S&P 500 and Nifty 50 weekly log returns. Run the Gibbs Sampler for 10,000 iterations (with 2,000 burn-in) and extract: (a) posterior distributions of $\mu_j$, $\sigma_j^2$, $p_{jj}$; (b) smoothed regime probabilities $P(S_t = 1 \mid \mathbf{y})$ over

the full sample. Validate by checking whether high-uncertainty periods align with known crises (2008, 2020).

## 6. Expected Results and Learning Outcomes

**Computational Results:**

- A fully functional Gibbs Sampler implemented from scratch in Python.
- Trace plots and ACF plots demonstrating sampler convergence.
- Gelman-Rubin $\hat{R}$ statistics near 1 for all parameters.

**Empirical Results:**

- Recovered posterior distributions for regime means and variances in S&P 500 and Nifty 50 return series.
- Smoothed probability plots identifying structural breaks consistent with the 2008 Financial Crisis and the 2020 COVID-19 shock.
- Comparison of Indian and US market regime dynamics.

## 7. Data Availability Statement

| **Dataset** | Source | Link / Access |
|---|---|---|
| **S&P 500 Returns** | Yahoo Finance | `pypi.org/project/yfinance` (Free API) |
| **Nifty 50 Data** | NSE India | `nseindia.com` (Free download) |
| **US GDP Growth** | FRED | `fred.stlouisfed.org/series/A191RL1Q225SBEA` |
| **Recession Indicator** | FRED (NBER) | `fred.stlouisfed.org/series/USREC` |
| **India GDP** | RBI DBIE | `dbie.rbi.org.in` (Free) |

All datasets used are **publicly available** through the repositories listed above and require no special access. We will not request primary data from paper authors, as the public repositories are sufficient. We will additionally use **synthetic data** simulated from the Markov Switching Model to validate the sampler before applying it to real data. We do not plan to collect primary data through surveys.

## 8. AI-Usage and Plagiarism Statement

Our group will utilize AI tools, including ChatGPT and Gemini, primarily for code optimization, LaTeX formatting, and structural brainstorming. To ensure orig- inality, all mathematical derivations of the full conditional distributions and the core implementation of the Gibbs Sampler will be developed from first principles by the group members. We will prevent plagiarism by using AI-generated suggestions only as a sec- ondary reference, ensuring that all final interpretations, empirical analysis of financial data, and written conclusions are our own unique work, cross-checked against academic sources and verified through standard plagiarism detection software.